Name of the Experiment CONSTRUCTION & GRAPHICAL REPRESENTATION OF FREQUENCY DISTRIBUTION

Date: 27.04.26
Exp. No.: 01
Page No.: 01

Problem 01: The following monthly incomes (100 Tk) of 64 farmers are given below:

20	24	03	13	15	21	08	02
04	19	16	22	24	23	14	22
15	07	26	27	19	19	23	18
17	32	18	30	28	24	06	23
23	14	11	22	14	29	03	20
12	36	29	18	12	22	04	13
23	03	31	26	09	16	13	24
28	08	19	15	06	09	24	21

a) Construct frequency distribution using suitable class interval as inclusive and exclusive method

b) Draw the following diagrams -

Bar Diagram, Histogram, frequency polygon, Frequency curve and ogive.

Name of the Experiment CONTINUED

Date: 27.04.26
Exp. No.: 01
Page No.: 02

(a) Solution :

Here, Total observation, N = 64

Highest value = 36

lowest value = 03

Range = Highest value - lowest value

= (36 - 03) = 33

From Sturge Rule,

$K = 1 + 3.222 \log_{10} N$

$= 1 + 3.222 \log_{10} 64$

$= 6.819$

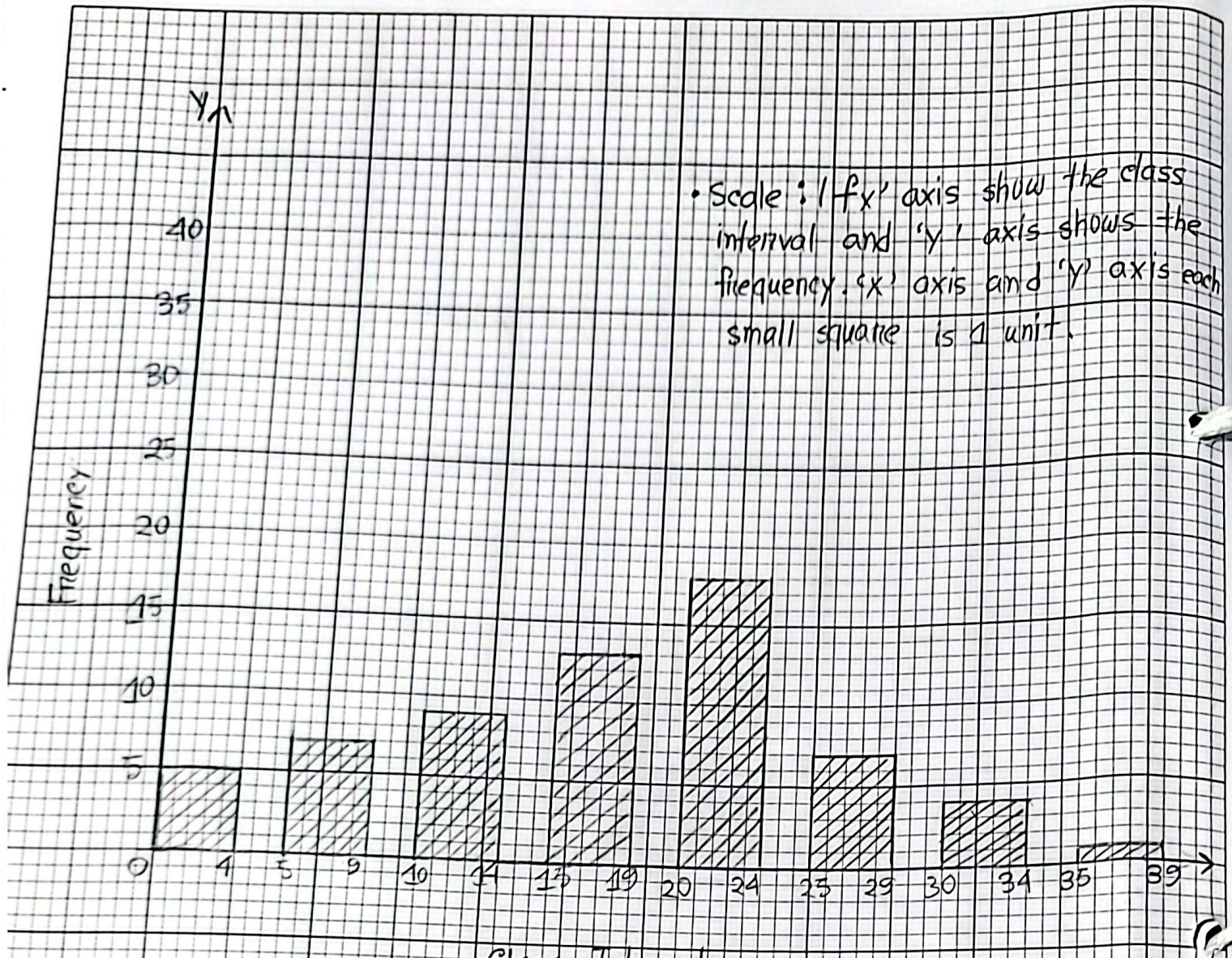
$\text{Class Interval (CI)} = \frac{\text{Range}}{\text{No. of classes}}$

$= \frac{33}{6.819} = 4.83$

∴ We can consider class interval as 5 because of the next round figure.

Signature

Figure No.: Page No.: Date:



• Scale: In 'x' axis show the class interval and 'y' axis shows the frequency. 'x' axis and 'y' axis each small square is 1 unit.

Fig: Bar Diagram

Comment: The bar diagram shows that highest frequency remain in class (20-24) and lowest frequency in class (35-39)

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

Page No.: 03

Inclusive:

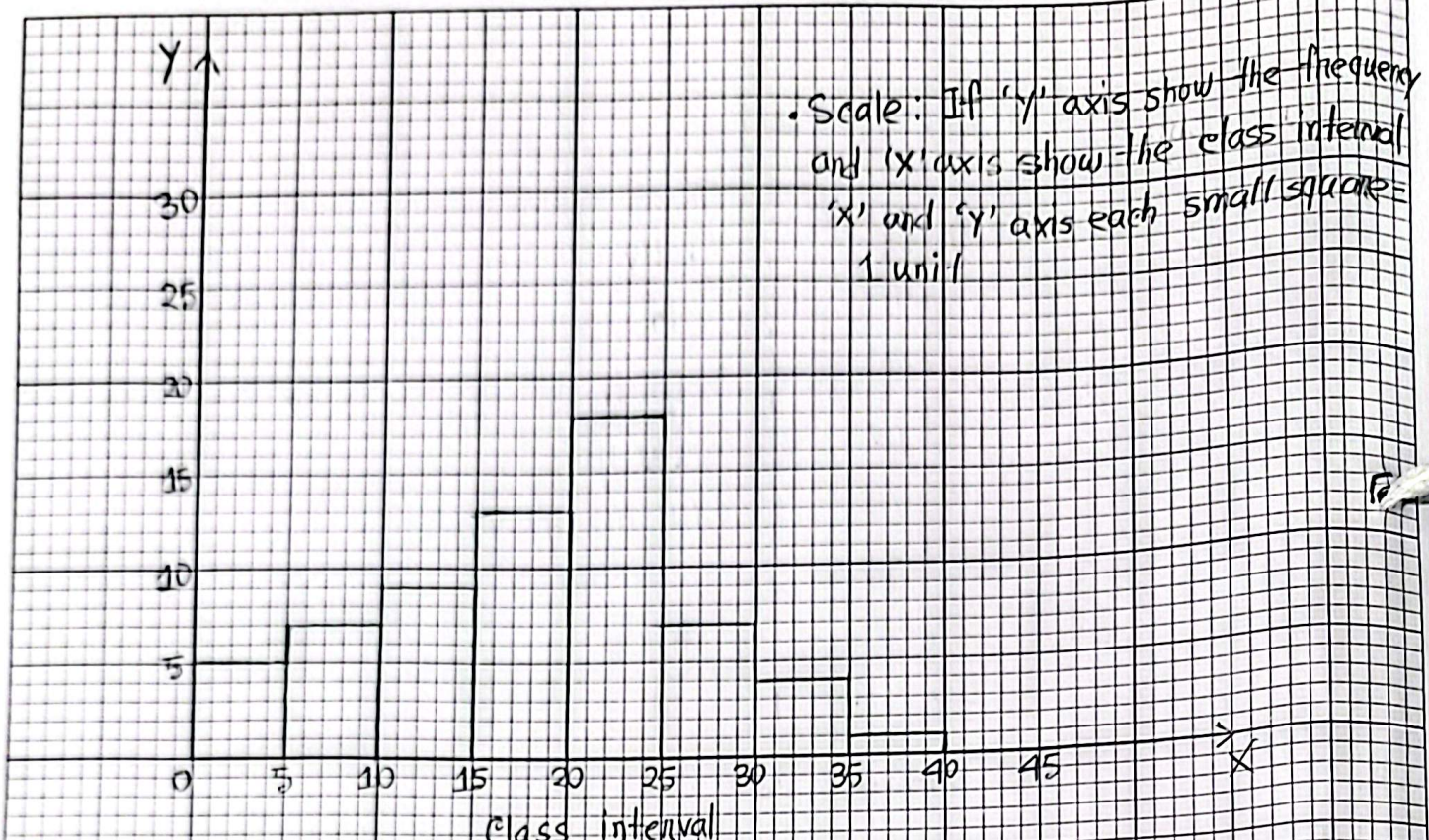
Class	Tally marks	Frequency
0 - 4	𝍸	5
5 - 9	𝍸 II	7
10 - 14	𝍸 IIII	9
15 - 19	𝍸 𝍸 III	13
20 - 24	𝍸 𝍸 𝍸 III	18
25 - 29	𝍸 II	7
30 - 34	IIII	4
35 - 39	I	1

Exclusive:

Class	Tally marks	Frequency	Mid value	Cumulative frequency
0 - 5	𝍸	5	2.5	5
5 - 10	𝍸 II	7	7.5	12
10 - 15	𝍸 IIII	9	12.5	21
15 - 20	𝍸 𝍸 III	13	17.5	34
20 - 25	𝍸 𝍸 𝍸 III	18	22.5	52
25 - 30	𝍸 II	7	27.5	59
30 - 35	IIII	4	32.5	63
35 - 40	I	1	37.5	64

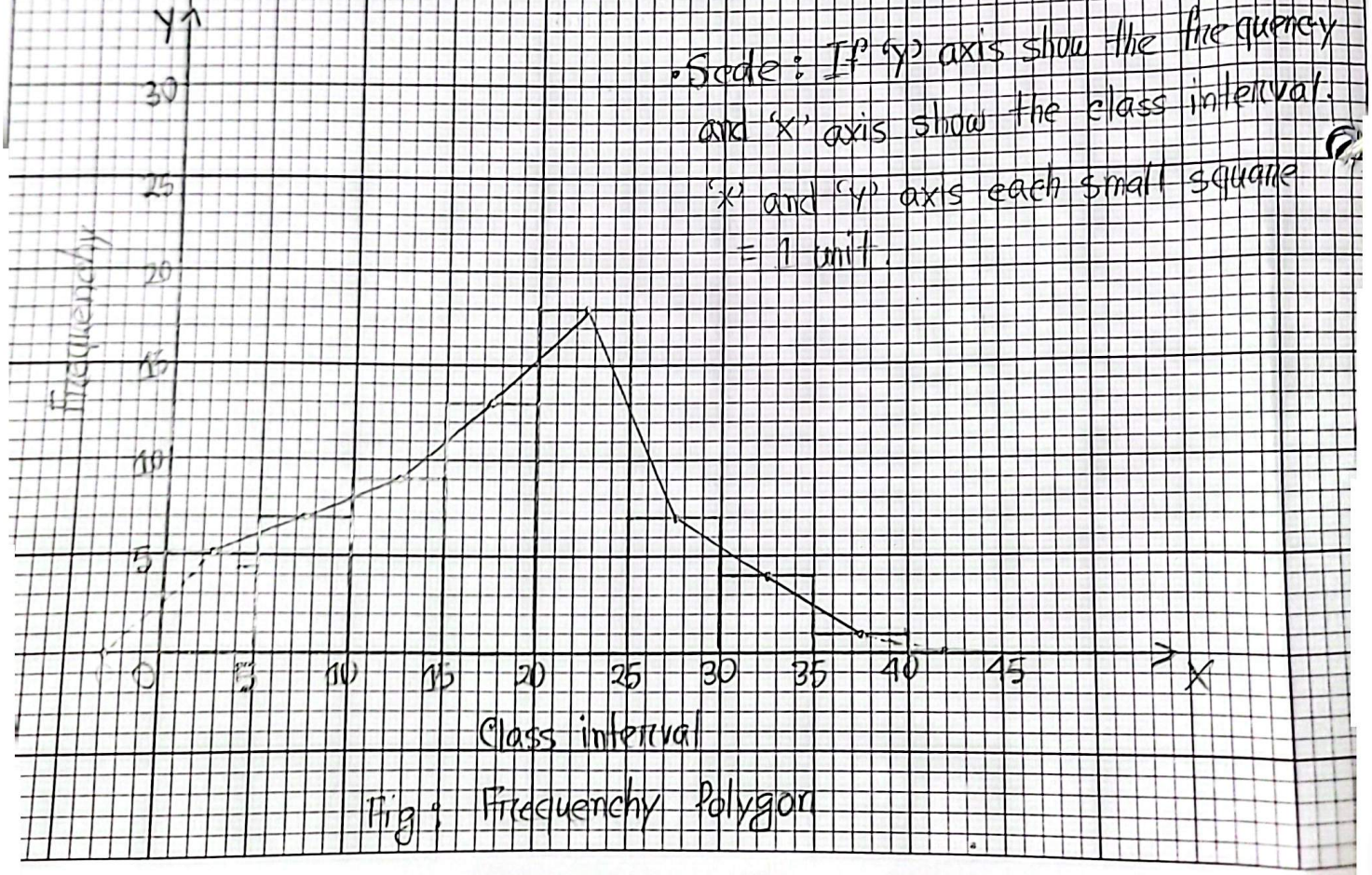
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Figure No.: Page No.: Date:



• Scale: If 'y' axis show the frequency and 'x' axis show the class interval 'x' and 'y' axis each small square = 1 unit

Fig: Histogram



• Scale: If 'y' axis show the frequency and 'x' axis show the class interval. 'x' and 'y' axis each small square = 1 unit.

Fig: Frequency Polygon

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

Page No.: 04

b) Histogram: In the graph paper x axis shows the class interval (CI) and each small square = 1 unit. The y axis shows the frequency and each small square = 1 unit. The histogram shows that highest frequency remain in class interval (20-25) and lowest remain in class interval (35-40).

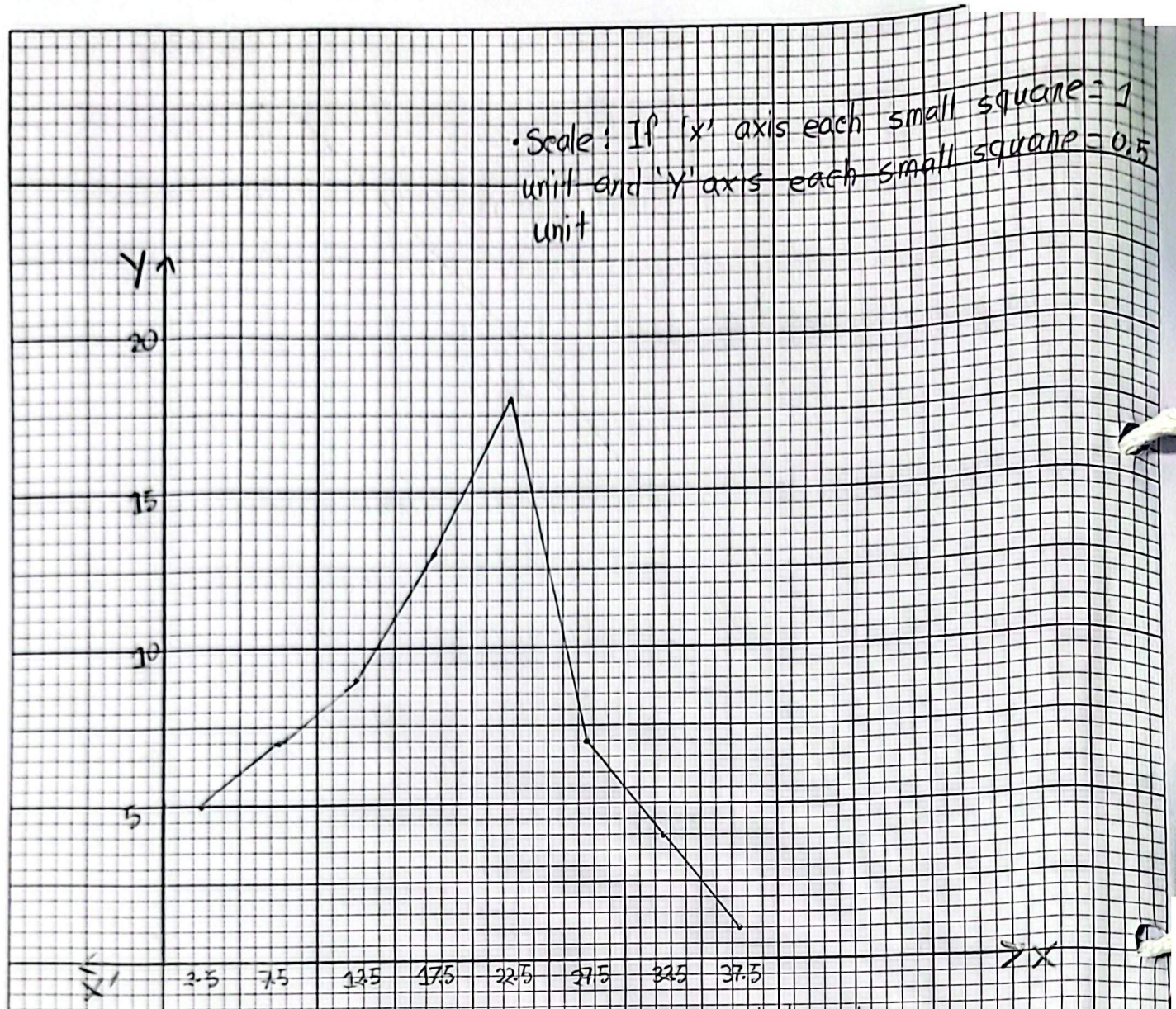
Frequency Polygon: In the graph paper, the x axis shows the class interval (CI) and each small square = 1 unit. The y axis shows the frequency and small square = 1 unit.

The upward rate of frequency polygon increase of first and becomes highest as class interval (20-25) and then gradually downward.

Signature

Figure No.: Page No.:

• Scale: If 'x' axis each small square = 1 unit and 'Y' axis each small square = 0.5 unit



Mid value of class interval

Fig: Frequency curve

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

Page No.: 05

Frequency curve: In the graph paper shows 'x' axis the mid value of class interval and each small square = 1 unit. The 'y' axis the frequency value are shown and each small square = 0.5 unit.

The upward rate of frequency curve increase at first and becomes highest at mid value of 22.5 and then gradually downward.

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Figure No.: Page No.: Date:

• Scale: If 'x' axis each small square = 1 unit and 'y' axis each small square = 2 unit.

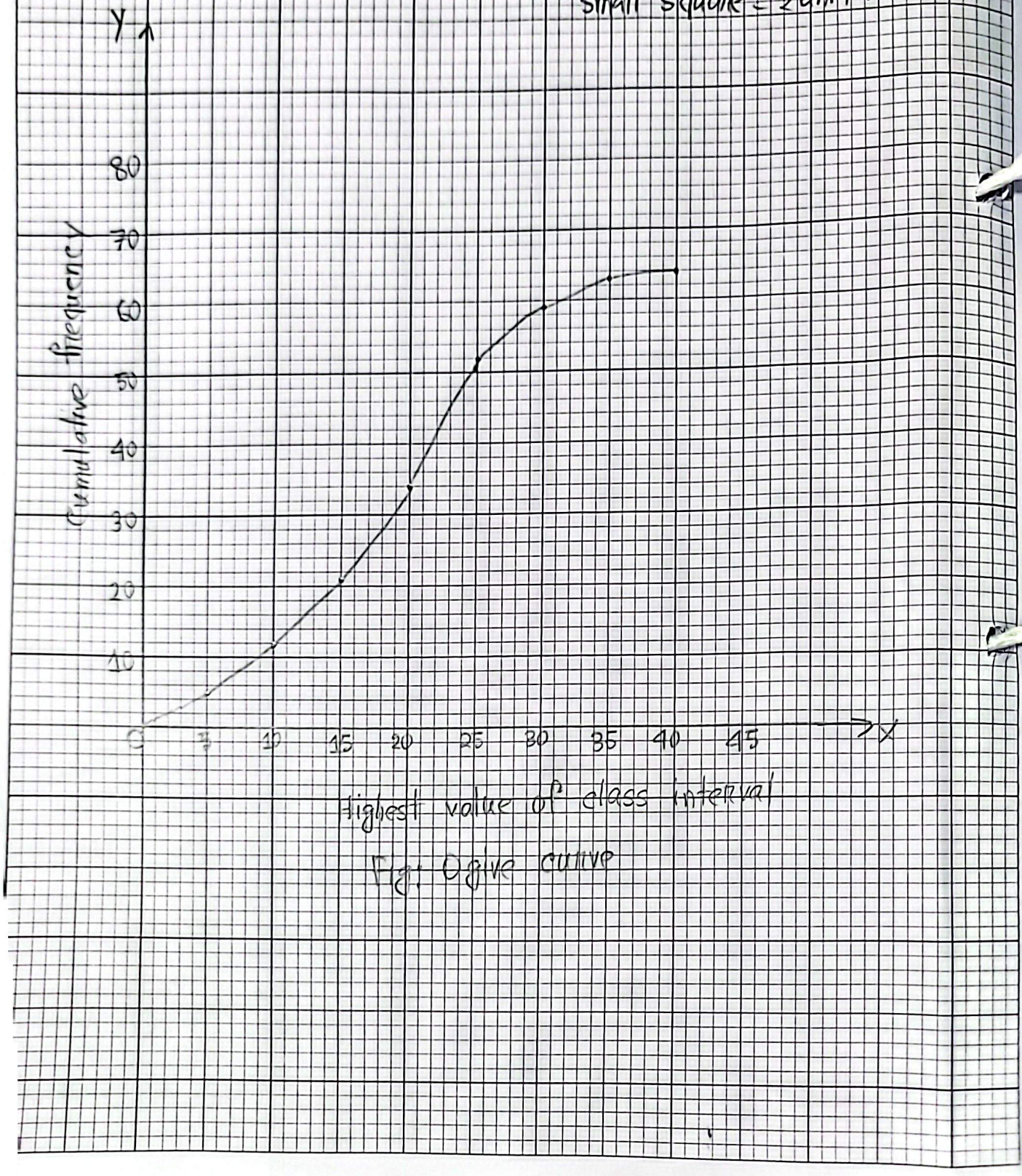


Fig: Ogive curve

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

Page No.: 06

Ogive curve / Cumulative frequency curve:

In the graph paper 'x' axis shows highest value of class interval and each small square = 1 unit. The 'y' axis shows the cumulative frequency and here each small square = 2 units. The upward rate of cumulative frequency / Ogive curve increase at first.

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Name of the Experiment CONTINUED

Date: 27.09.26
Exp. No.: 01
Page No.: 07

Problem 02:

The growth of population in three countries during the period (1901 - 1961) :

Year	Country A	Country B	Country C
1901	28.9	16.6	45.5
1911	31.6	19.4	50.9
1921	33.3	21.1	54.4
1931	35.6	23.6	59.2
1941	40.0	28.3	70.3
1951	42.1	33.7	75.6
1961	50.8	42.9	93.7

a) Construct a simple bar diagram for country A

b) Draw the multiple bar diagram using the growth of population of country B and C.

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Figure No.: Page No.: Date:

- Scale: 'X' axis show the year and 'Y' axis show the population. In 'Y' axis each small square = 2 unit

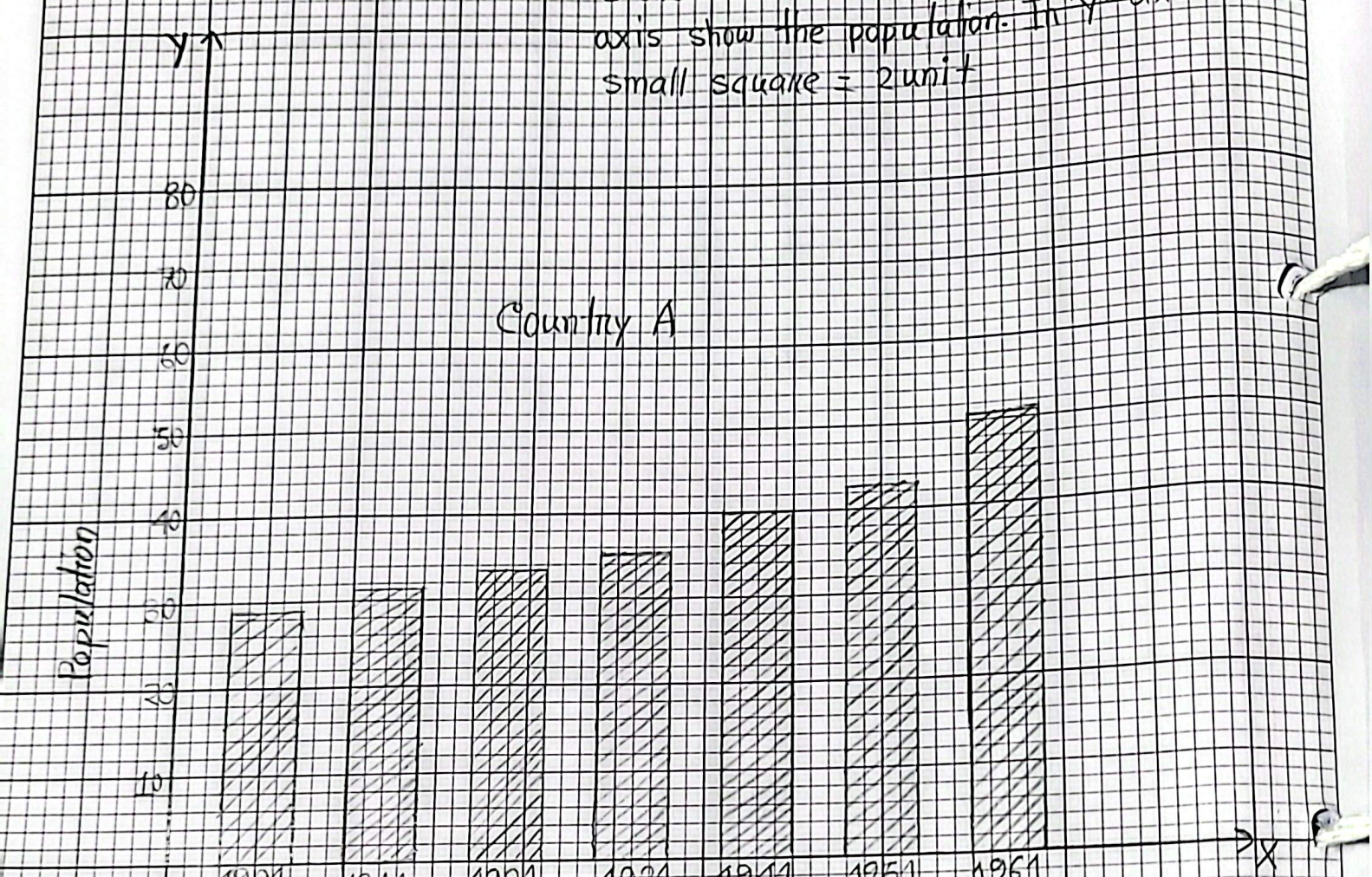


Fig: Single Bar diagram

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

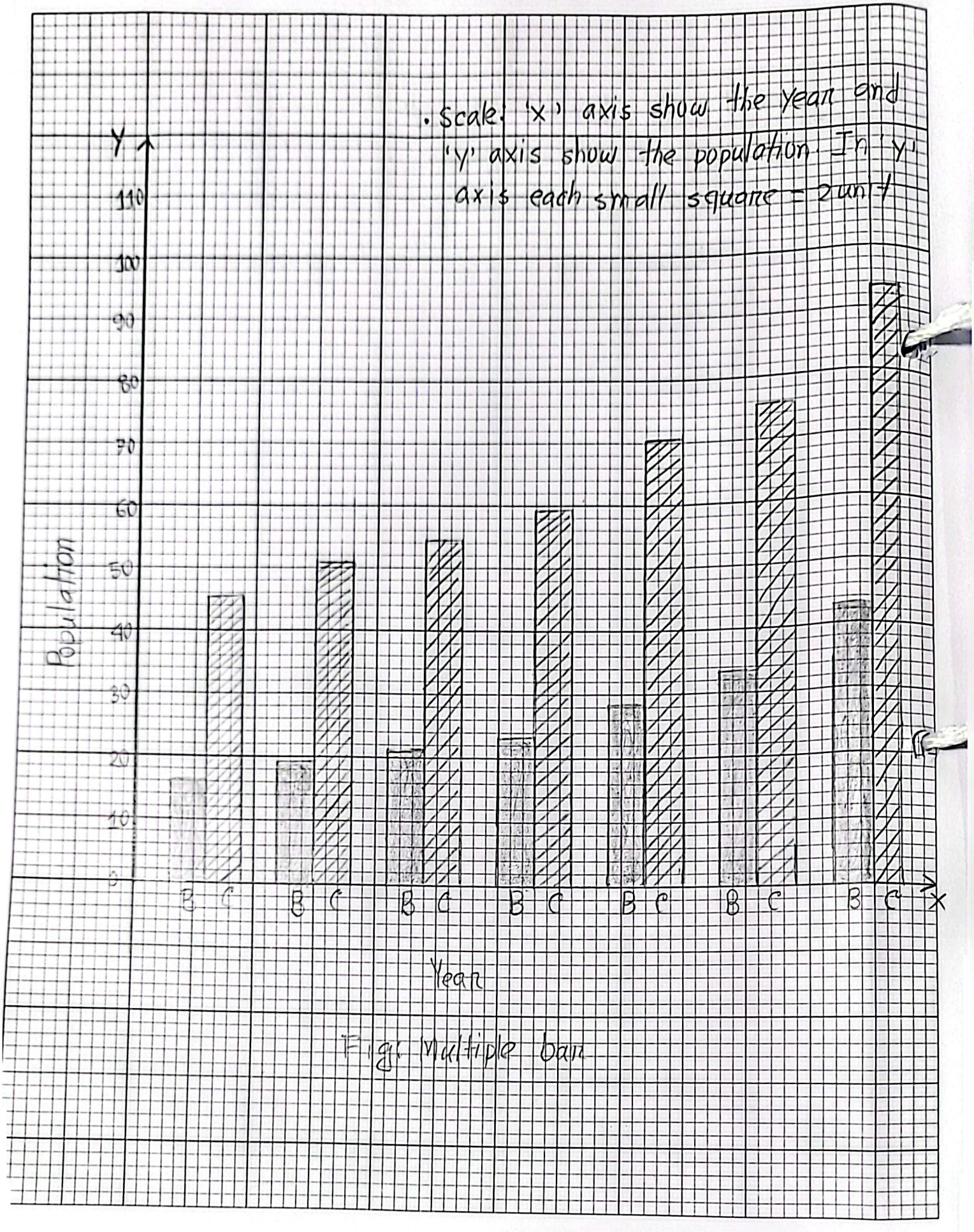
Page No.: 08

Comment for A: From the simple bar diagram we can see the growth of population rate of country A was lowest in 1901 and highest in 1961 respectively. The growth of population was increasing day by day.

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Figure No.. Page No.: Date:



. Scale: 'x' axis show the year and 'y' axis show the population. In 'y' axis each small square = 2unit

Fig: Multiple bar

Name of the Experiment CONTINUED

Date: 27.04.26

Exp. No.: 01

Page No.: 09

Comment for B & C: From the multiple bar diagram we can compare that the growth of population rate between country B and C. In country C the growth rate higher than B. But in both country, thy population was increasing day by day

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Figure No.: Page No.: Date:

• Scale: 'x' axis shows the country
'y' axis shows the area. In 'y' axis
each small area = 0.5 unit

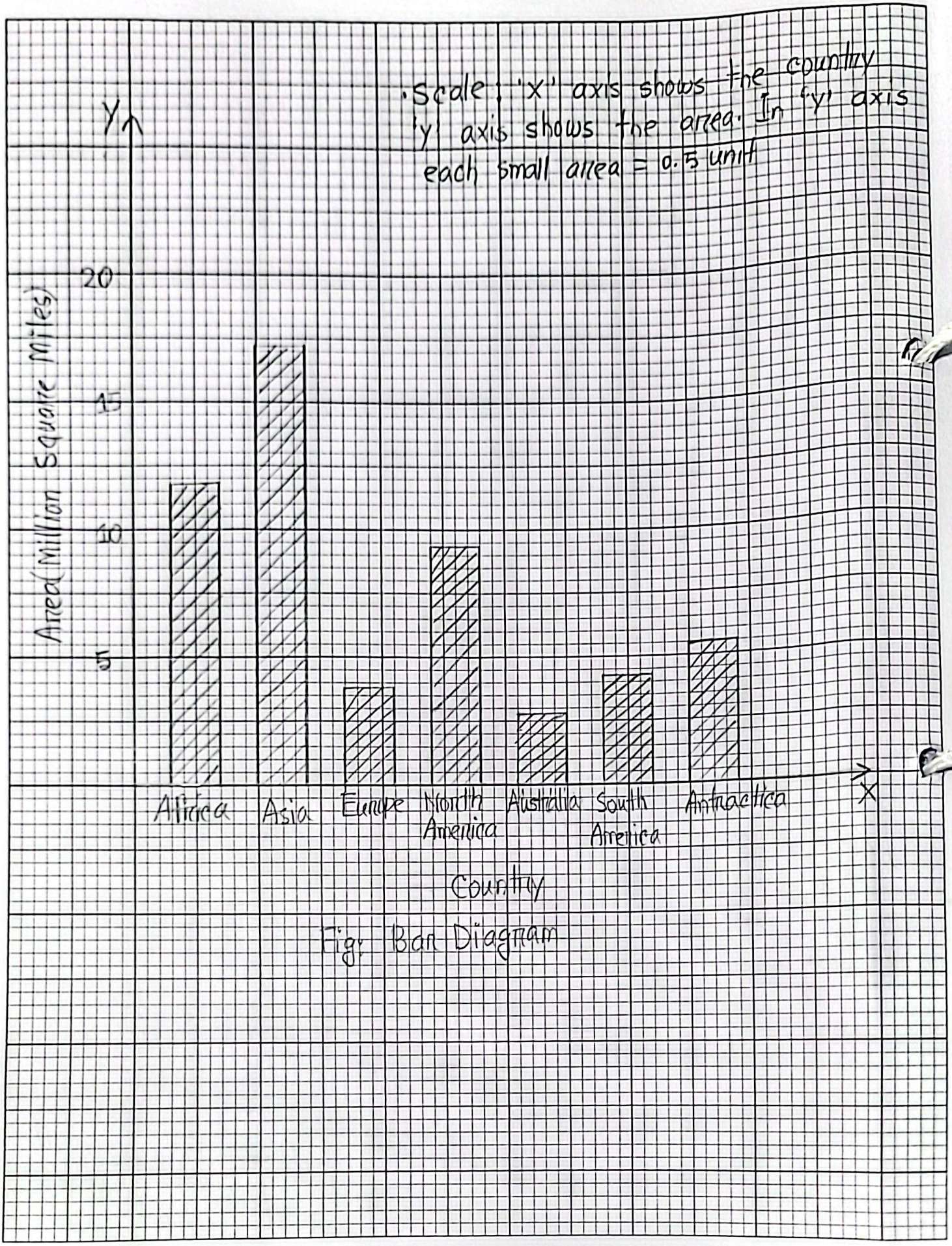


Fig: Bar Diagram

Problem 03:

The area of continents of the world is given below:

Continent	Area (Million sq miles)
Africa	11.7
Asia	17.1
Europe	3.9
North America	9.4
Australia(Ocenia)	2.9
South America	6.9
Antarctica	5.5

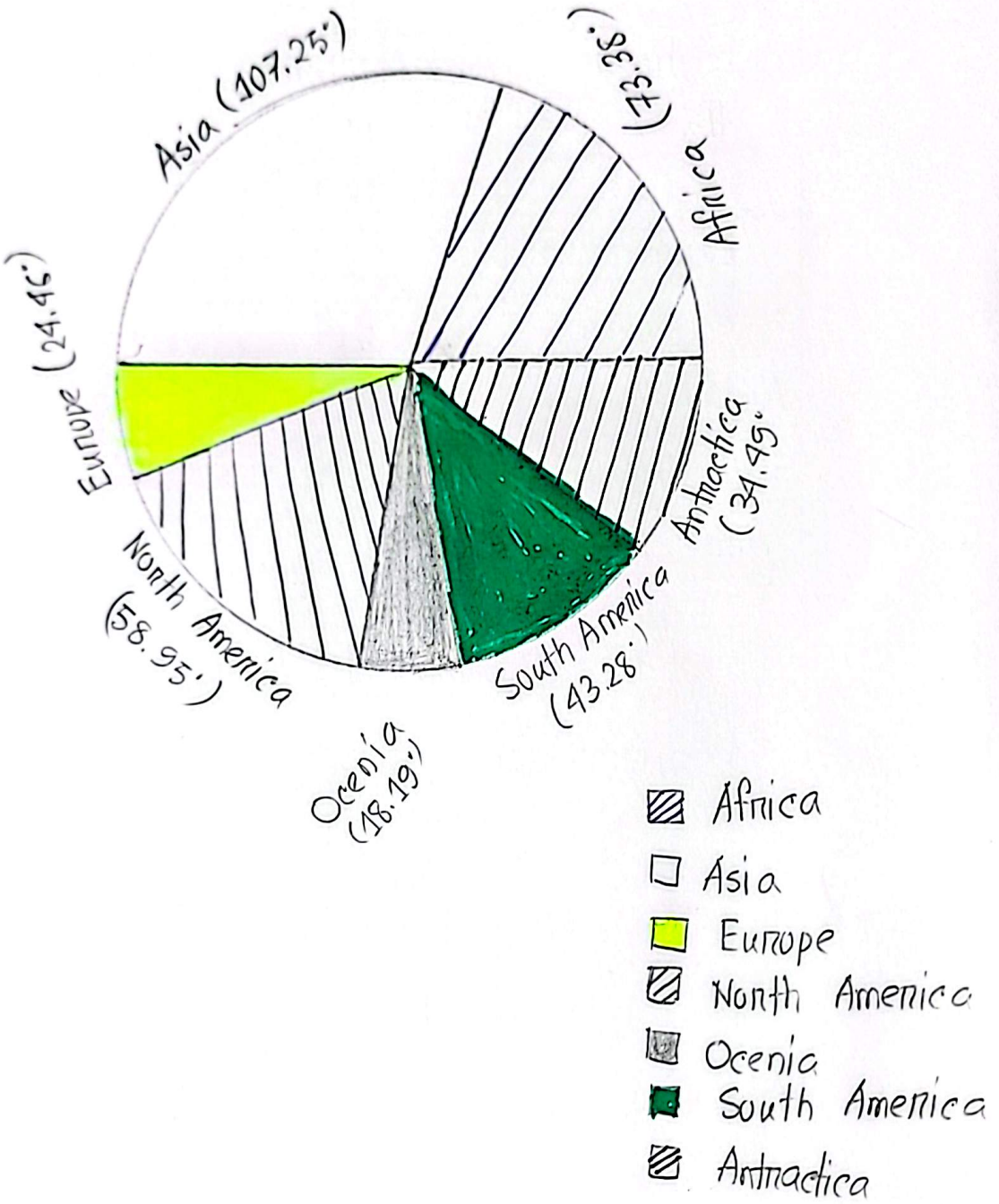
a) Draw a bar diagram using the table

b) Draw a pie chart

a) Comment: From the bar diagram we can see that the area for Asia is the highest and the Area of Australia is the lowest than the other continents.

Signature

Figure No.: Page No.: Date:



Name of the Experiment CONSTRUCTION & GRAPHICAL REPRESENTATION OF FREQUENCY DISTRIBUTION

Date: 27.04.26

Exp. No.: 01

Page No.: 11

b)

Continent	Area (Millions square miles)	Angle of area
Africa	11.7	$\frac{11.7}{57.4} \times 360^\circ = 73.38^\circ$
Asia	17.1	$\frac{17.1}{57.4} \times 360^\circ = 107.25^\circ$
Europe	3.9	$\frac{3.9}{57.4} \times 360^\circ = 24.46^\circ$
North America	9.4	$\frac{9.4}{57.4} \times 360^\circ = 58.95^\circ$
Australia (Oceania)	2.9	$\frac{2.9}{57.4} \times 360^\circ = 18.19^\circ$
South America	6.9	$\frac{6.9}{57.4} \times 360^\circ = 43.28^\circ$
Antractica	5.5	$\frac{5.5}{57.4} \times 360^\circ = 34.49^\circ$

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